

ERES-BERA-ODOR-2026-001 v1.0

ODOR as ARI Sensor Substrate

A BERA / FAVORS Addendum — ERES Institute for New Age Cybernetics

Document Control

Designation: ERES-BERA-ODOR-2026-001

Version: v1.0 — DRAFT (HYPOTHESIZED tier)

Date: June 19, 2026

Author: Joseph Allen Sprute (JAS), Founder & Director, ERES Institute for New Age Cybernetics

Parent Instruments: FAVORS (Fingerprint·Aura·Voice·ODOR·Retina·Signature); BERA Measurement Specification v0.2.1; Aura Resonance Index (ARI)

Status: Pending MIEVM ensemble review

1. Purpose

This addendum formalizes ODOR as the physical sensor substrate for the Aura Resonance Index (ARI), converting ARI from an architecturally named index into one with an actual chemical instrumentation pathway. It does not replace or redefine Aura conceptually — it gives the concept a body.

2. Grounding — Why ODOR

FAVORS splits structurally into two classes. Static-identity markers confirm who, against a fixed reference. State-resonance markers report a condition — consistent with the BERA Notation Lock’s definition of resonance as a measurable energetic state of a person, a state, never a standing.

Marker Class	FAVORS Elements	Behavior Under Conscious Control?
Static Identity	Fingerprint · Retina · Signature	N/A — fixed reference, not state-dependent
State-Resonance	Aura · Voice	Voice: partially — can be performed/rehearsed
State-Resonance (involuntary)	ODOR	No — VOC emission shifts with cortisol/adrenaline outside conscious control

ODOR is the least consciously controllable of the six elements. Cortisol- and adrenaline-linked volatile organic compounds (VOCs) in breath and sweat shift involuntarily with stress, fear, attraction, and arousal — underneath conscious performance in a way voice, posture, or expression are not.

3. The Substrate Claim

ARI has, until now, lacked a direct physical readout — “Aura” as a term risked remaining poetic rather than instrumented. This addendum proposes: ARI’s primary instrumentable channel is the wearer’s VOC emission profile, captured via a chemical (e-nose / gas) sensor.

4. Hardware Precedent (Non-Hypothetical)

This is not a speculative leap. Research-grade smart-glasses platforms already exist, instrumented for exactly this purpose:

- AirSpec — a smart-glasses sensing platform built for monitoring personal and built-environment chemical exposure, presented in academic wearable-computing research (ACM UbiComp/ISWC adjunct proceedings, 2023).
- A companion Affective Air Quality dataset pairs personal chemical emissions with emotional-state video data — establishing an empirical link between VOC signal and affective/stress state, which is exactly what ARI requires to move from concept to measurement.

Placement rationale: the nose-bridge — where smart glasses already sit on every wearer — sits in the exhaled-breath plume and close to skin-volatile emission zones. No design compromise is required to add a chemical sensor to a wearable form factor already worn for unrelated reasons (display, camera, audio).

Consumer-market caveat: as of mid-2026, no mainstream consumer smart glasses (Ray-Ban Meta, Snap Specs, Even Realities, and similar) ship with a chemical/VOC sensor. The instrumentation pathway exists at research-prototype maturity, not consumer-shipped maturity. This addendum’s status reflects that gap honestly — ARI-via-ODOR is one hardware generation away, not deployed.

5. Scope Limit

This addendum applies to in-person, sensor-proximate BERA/FAVORS instrumentation (e.g., wearable verification contexts). It does not extend to remote or voice-only instruments — EEP Layer 1 (ERES-EEP-2026-001, the crisis-line number) has no ODOR channel and is unaffected by anything in this addendum.

6. Open Engineering Questions (Pre-MIEVM)

- Sensor specificity — which VOC species map to which ARI sub-states (stress vs. attraction vs. baseline), and at what false-positive / false-negative rate.
- Consent and privacy layer — chemical emission is involuntary; the consent model must be explicit and revocable independent of the sensor’s physical presence.

- Calibration drift — baseline VOC profiles vary by diet, medication, and health status; ARI normalization must account for this or risk misreading a medical state as a resonance state.
- Integration point — where in the existing ARI formula a VOC-derived signal enters: as ARI itself, or as one weighted input among several.

7. Status & Next Steps

Status: HYPOTHESIZED tier. This addendum requires:

- MIEVM ensemble review (Claude, Grok, DeepSeek, ChatGPT; 3-of-4 concurrence, GO threshold ≥ 8.5).
- A named hardware or research partner contact (e.g., the AirSpec research lineage) before this addendum is treated as anything beyond conceptual.
- JAS confirmation that ARI's chemical substrate claim is intended as literal instrumentation, not metaphorical architecture.

8. References

AirSpec: A Smart Glasses Platform, Tailored for Research in the Built Environment. ACM UbiComp/ISWC 2023 Adjunct Proceedings.

Affective Air Quality Dataset: Personal Chemical Emissions from Emotional Videos. arXiv preprint, 2025.